

HAMZA MANZOOR

Senior Full-Stack Engineer · Generative AI Integration · LLM-Powered SaaS

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PROFESSIONAL SUMMARY

Senior Full-Stack Engineer with 6+ years of experience designing, building, and shipping production-grade SaaS platforms — with a current, hands-on focus on Generative AI integration and LLM-powered product features. Specialise in event-driven backend architecture (NestJS / Node.js / FastAPI), React / Next.js frontends, and AI orchestration pipelines using OpenAI GPT-4o, Anthropic Claude, and Google Gemini APIs. Currently at Voxbee, engineering the full AI media generation stack: multi-model orchestration, AI dubbing and TTS pipelines, GPU inference containers on AWS, BullMQ job queuing with crash-safe reconciliation, and direct-to-S3 multi-tenant file management. Comfortable owning features end-to-end — from system design and database modelling to containerised deployment and production monitoring. Delivered production systems for US, UK, EU, and Pakistani clients across marketing analytics, hospitality, enterprise recruitment, salon management, and creator-economy verticals. Strong communicator; at ease working with product managers, designers, and cross-functional remote teams in Agile environments.

TECHNICAL SKILLS

Generative AI & LLMs: OpenAI GPT-4o / GPT-4 Turbo / GPT-3.5 · Anthropic Claude 3.5 Sonnet / Claude API · Google Gemini 1.5 Pro · LLM chaining & multi-step orchestration · Prompt engineering (system prompts, few-shot, chain-of-thought) · RAG pipelines (chunking, embedding, vector retrieval) · OpenAI Embeddings / Pinecone / pgvector · AI dubbing & TTS (ElevenLabs, custom pipelines) · Stable Diffusion · DALL-E 3 · Image generation APIs · Model quantisation (GPTQ / GGUF, 4-bit / 8-bit) · vLLM inference server · ONNX Runtime · PyTorch / TensorFlow inference · LoRA / QLoRA fine-tuning concepts · AWS SageMaker endpoints

Backend & APIs: NestJS (modules, guards, interceptors, pipes, decorators) · Node.js (event loop, streams, clustering) · Express.js · FastAPI (Python, async) · REST API design · GraphQL (basic) · WebSockets (Socket.io) · BullMQ / Bull job queues · JWT · Passport.js · RBAC · Stripe · Multi-gateway payment integration · Webhook design · Swagger / OpenAPI · Microservices

Frontend: React.js (hooks, context, performance optimisation) · Next.js (App Router, SSR, SSG, ISR) · TypeScript · Redux / Redux Saga / Redux Toolkit · Tailwind CSS · Ant Design · Material UI · Bootstrap · Styled Components · SCSS · GSAP animations · D3.js · Storybook · Figma · Adobe XD · Responsive / mobile-first design · Web performance (Core Web Vitals)

Databases: PostgreSQL (schema design, indexing, query optimisation, CTEs, transactions, Sequelize ORM, Baseline + Delta migration strategy) · MongoDB (aggregation pipelines, indexing, Mongoose) · Redis (pub/sub, caching, BullMQ backend) · Firebase (Firestore, Auth, Cloud Messaging)

Cloud & DevOps: AWS (EC2, S3 + presigned URLs, RDS, Lambda, SageMaker, CloudFront, IAM, VPC basics) · Docker · Docker Compose · Nginx (reverse proxy, SSL, load balancing) · GitHub Actions CI/CD · Linux (Ubuntu, shell scripting) · CUDA 12.x (GPU inference containers)

Mobile: React Native · Expo · Firebase Auth & Firestore · Push notifications · Google Play Store deployment

Testing & Tooling: Playwright (E2E testing & web scraping) · Jest (unit testing basics) · Git · GitHub · Postman · Clean Architecture · Factory & Repository Pattern · Agile / Scrum

WORK EXPERIENCE

Senior Full-Stack Engineer — Generative AI Platform · Voxbee (Xccelerated.ai)

Jan 2026 – Present

AI-powered media generation SaaS covering dubbing, TTS, voice cloning, AI avatars, image/video synthesis, music, and podcast production.

- ▶ **Multi-model AI orchestration:** Designed and maintained a MODEL_SERVICE_MAP routing layer that dispatches generation requests to the correct AI model — internal GPU workers or third-party APIs — based on media type,

quality tier, and resource availability. Enables hot-swapping of models (e.g. switching TTS providers or music generation backends) without service restarts or code deploys.

- ▶ **LLM API integration:** Integrated OpenAI GPT-4o, Anthropic Claude 3.5 Sonnet, and Google Gemini 1.5 Pro APIs for scriptwriting, translation, content summarisation, and dubbing script adaptation within the platform. Designed reusable prompt templates and multi-step LLM chains for context-aware media localisation workflows.
- ▶ **Direct-to-S3 file management module:** Architected a multi-tenant file management system using Factory + Repository Pattern (TypeScript / Express). Backend acts as a security orchestrator: validates tenant context, generates short-lived presigned S3 upload URLs, tracks file metadata and lifecycle state in PostgreSQL, and enforces per-tenant isolation — preventing any cross-tenant data access.
- ▶ **Distributed AI job queue with crash recovery:** Built a BullMQ-based job queue for AI generation tasks (audio rendering, video synthesis, music generation) running across NestJS and FastAPI workers. Implemented a reconciliation routine that runs on every worker restart: queries PostgreSQL for jobs in 'processing' state older than a threshold, requeues them with idempotency keys, and marks ghost jobs as failed — eliminating duplicate processing and stuck-job accumulation.
- ▶ **GPU inference container stack:** Resolved deep CUDA / Docker compatibility issues in the mini_worker service — the audio and speech processing container running TensorFlow, PyTorch, and ONNX Runtime simultaneously. Aligned all framework CUDA dependency versions against CUDA 12.x constraints, fixed cuDNN library conflicts, and produced a stable multi-framework GPU inference image deployable on AWS EC2 GPU instances.
- ▶ **AI dubbing & TTS pipeline:** Contributed to the dubbing pipeline: transcript ingestion, source-language detection, LLM-based script adaptation for target language (preserving lip-sync timing constraints), TTS synthesis via ElevenLabs and custom voice models, and final audio track muxing with the source video asset stored in S3.
- ▶ **AI music generation (ACEStep) & podcast (Podcastfy):** Integrated internal ACEStep music generation model and Podcastfy podcast production pipeline into the orchestration layer; built the async job lifecycle management, progress callbacks, and output delivery to the frontend via WebSocket and polling endpoints.
- ▶ Collaborated on avatar synthesis and video generation workflows, handling large binary asset handoff between GPU workers and S3/CDN delivery layers.

Senior Full-Stack Developer · Tech Savvy Startup — SalonX

Oct 2025 – Dec 2025

Multi-tenant white-label salon booking SaaS: each partner salon gets an embeddable booking widget for their own website.

- ▶ **Multi-tenant architecture:** Designed a NestJS + PostgreSQL multi-tenant data model with strict row-level tenant isolation. Each salon account has independent scheduling rules, service catalogues, staff profiles, pricing, and customer data — all scoped by tenantId enforced at the ORM and middleware layer.
- ▶ **Embeddable booking widget:** Built per-salon booking widgets as Next.js micro-frontends with server-side rendering for fast initial load and SEO on partner salon websites. Widgets load via a single script tag, resolve the tenant from a public API key, and render a fully branded booking flow without exposing internal infrastructure.
- ▶ **Payment gateway integration:** Integrated Stripe and local payment providers with webhook-driven order reconciliation — handling payment capture, partial refunds, appointment cancellation credits, and retry logic for failed webhooks.
- ▶ **Real-time appointment system:** Implemented WebSocket-based real-time notifications (Socket.io) for appointment state changes — instant updates to salon staff dashboards when customers book, reschedule, or cancel.
- ▶ **Staff scheduling engine:** Built a scheduling engine supporting per-staff availability windows, blocked times, service duration constraints, and buffer time between appointments — preventing double-booking across concurrent requests using PostgreSQL advisory locks.
- ▶ CI/CD pipeline: GitHub Actions deploying to AWS EC2 with Docker Compose, Nginx reverse proxy, SSL termination via Let's Encrypt, and zero-downtime rolling deploys.
- ▶ RBAC with four roles: Salon Owner (full control), Manager (operations), Staff (own schedule), Customer (booking only) — enforced via NestJS Guards and custom decorators.

Senior Full-Stack Developer · Techmate Tech LLC

Oct 2024 – Sep 2025

- ▶ **Marketing.biz — AI marketing analytics platform:** Delivered a multi-channel social analytics and content platform aggregating data from Facebook Graph API, Instagram Basic Display API, LinkedIn Marketing API, YouTube Data API v3, and Twitter/X API v2. Unified performance metrics — reach, impressions, engagement rate, follower growth — into per-client dashboards with time-series trend analysis and cross-platform benchmarking.
- ▶ **AI content generation pipeline:** Integrated GPT-4 for AI-driven post copywriting (with tone, platform, and audience controls) and DALL·E 3 / Stable Diffusion for AI image generation — enabling marketers to create and schedule optimised content without leaving the platform. Designed prompt templates that adapt output style per social channel (LinkedIn professional tone vs Instagram casual).
- ▶ **Per-client async analytics engine:** Engineered isolated data ingestion pipelines per client account using BullMQ queues with priority tiers. Redis caching with per-platform TTLs ensured API rate-limit compliance across five platforms simultaneously. Background refresh jobs kept dashboards near-real-time without blocking user requests.
- ▶ **Romingo — hotel management system:** Inherited an existing codebase and performed a full architecture review; refactored service layer for separation of concerns and testability. Rewrote N+1 heavy Sequelize ORM queries using proper eager loading and raw SQL where needed. Introduced PostgreSQL composite indexes on high-traffic query paths, significantly reducing average API response times. Added Redis caching for room availability lookups.
- ▶ **Database management:** Designed and optimised MongoDB aggregation pipelines for analytics data stores and managed PostgreSQL schema migrations, indexing strategies, and query plans across multiple production applications.
- ▶ **AWS infrastructure ownership:** Managed EC2 instance provisioning and hardening, RDS PostgreSQL administration, S3 bucket policies, CloudFront CDN configuration, IAM roles and least-privilege policies, Docker containerisation, and automated GitHub Actions CI/CD pipelines for all production deployments.
- ▶ Authored Swagger / OpenAPI documentation for all internal and client-facing API endpoints; maintained a Postman collection for QA and integration testing.

Senior Frontend Developer · Xeurix

Jul 2021 – Sep 2024

xeurix.com — Large-scale enterprise recruitment and workflow management platform serving enterprise clients.

- ▶ **Frontend architecture:** Led the frontend of a complex, multi-role enterprise recruitment SaaS from early MVP through to a production system with thousands of records. Owned component architecture decisions, state management strategy (Redux with Redux Toolkit), and styling system (Styled Components + Ant Design + Bootstrap).
- ▶ **Candidate pipeline UI:** Designed and built a multi-stage Kanban pipeline for candidate tracking — with drag-and-drop stage transitions, bulk action toolbars, advanced filter panels (by skills, location, status, date), saved filter presets, and real-time status updates via WebSocket integration.
- ▶ **Role-based dashboard system:** Engineered distinct dashboard views for Admin, Recruiter, Hiring Manager, and Candidate roles — each with tailored data visualisations (D3 charts for pipeline conversion rates, offer acceptance trends), notification centres, and action permissions enforced on both frontend and API boundary.
- ▶ **Performance optimisation:** Identified and resolved critical UI bottlenecks in large candidate list views — implemented windowed list rendering (react-window), memoised expensive selector computations with reselect, lazy-loaded heavy dashboard panels, and reduced initial bundle size via code splitting. Resulted in measurably faster load times on low-bandwidth connections.
- ▶ **API integration & collaboration:** Worked closely with backend engineers to define REST API contracts, shape pagination and filtering query parameters, and design error response formats — reducing integration friction across teams.
- ▶ **Agile delivery:** Participated in two-week Agile sprints: backlog refinement, sprint planning, daily standups, code reviews (PR feedback on patterns, accessibility, and performance), and retrospectives. Mentored two junior frontend developers on React patterns, Git workflow, and component design.
- ▶ Maintained reusable component library covering forms, modals, tables, and notification components — reducing duplication and accelerating feature delivery across quarters.

Junior JavaScript Developer · Xacrosoft

Sep 2019 – Jun 2021

- ▶ Built and maintained client web applications using React.js, Node.js, and Express.js across the full SDLC — from requirements and UI implementation to API integration, testing, and deployment.

- ▶ Implemented RESTful API integrations, JWT authentication flows, and form validation logic on the frontend and backend simultaneously — gaining full-stack fluency early in career.
- ▶ Contributed to database schema design decisions (MySQL / MongoDB), wrote seed scripts and migration files, and participated in production deployment procedures on shared hosting and early cloud environments.
- ▶ Collaborated in a small team with weekly code reviews, improving code quality habits around naming conventions, DRY principles, and error handling.

Frontend Developer Intern · Stepup Solution Provider

Jan 2018 – Aug 2018

- ▶ Gained hands-on proficiency in JavaScript (ES6+), HTML5, CSS3, React.js, and Bootstrap by contributing to real client projects under senior developer supervision.
- ▶ Built responsive landing pages and UI components; learned cross-browser compatibility, Git version control, and professional code review workflows.

FREELANCE & INDEPENDENT CONSULTING

Active freelance practice running alongside primary employment — delivering full-stack development, technical architecture consulting, SRS documentation, and client proposals for UK, US, Greek, and Pakistani clients. Fiverr gigs in full-stack development (NestJS backend, React frontend) and AI API integration (OpenAI, Claude, Gemini) — with production projects including Marketing.biz, SalonX, Romingo, Camperoni, and Xeurix.

- ▶ **Voxbee — S3 Infrastructure Roadmap:** Authored a detailed technical roadmap for migrating to a direct-to-S3 upload architecture: presigned URL flows, tenant isolation strategy, file lifecycle state machine, backend security orchestrator pattern, and PostgreSQL metadata schema design. Served as the implementation blueprint for the production build.
- ▶ **Imperium Executive Search Limited (UK):** Delivered a comprehensive website improvement audit and full redesign proposal for a London-based executive recruitment firm — covering UX/UI deficiencies, technical performance issues, SEO gaps, mobile responsiveness failures, and a phased redesign roadmap with developer handoff specifications.
- ▶ **Zain Sherazi Motor Market — Container Business System:** Produced a full SRS and client proposal for a container import business management platform: inventory and shipment tracking, customs clearance workflow, client portal with document uploads, financial reporting, and role-based access for operations staff.
- ▶ **Web Scrapers (Playwright / Python & JS):** Built production-grade scrapers for promptlibrary.space and BasedLabs using Playwright — handling dynamic JavaScript-rendered content, multi-page pagination, session-based authentication flows, rate-limit evasion, and structured JSON data extraction to PostgreSQL.
- ▶ **Real Estate Mobile Platform (Pakistan):** Authored SRS and technical proposal for a React Native real estate app: property listing with advanced filters, agent portal, appointment scheduling, Google Maps integration, mortgage calculator, and Firebase push notifications for new listings.
- ▶ **PostgreSQL DBA Consulting:** Produced a step-by-step PostgreSQL restore and disaster recovery guide for production databases; advised on Sequelize ORM Baseline + Delta migration strategy for applying schema changes to existing live data safely without downtime.

SELECTED PROJECTS

Voxbee AI Pipelines — Generative Media Platform — voxbee.com

- ▶ Multi-model orchestration (MODEL_SERVICE_MAP) routing AI generation tasks across OpenAI, Claude, Gemini, ElevenLabs TTS, ACEStep music generation, and Podcastfy podcast pipelines
- ▶ BullMQ job queue with PostgreSQL-backed crash recovery; GPU inference containers on AWS EC2 (CUDA 12.x); direct-to-S3 multi-tenant file management with presigned URL security orchestration
- ▶ Stack: NestJS · FastAPI · TypeScript · Python · BullMQ · PostgreSQL · Redis · Docker · AWS EC2/S3/SageMaker

Marketing.biz — AI Marketing Analytics Platform

- ▶ Aggregates 5 social platform APIs; AI post generation (GPT-4) + AI image creation (DALL-E 3 / Stable Diffusion); per-client BullMQ async pipelines with Redis rate-limit caching

- ▶ Stack: NestJS · React · PostgreSQL · MongoDB · Redis · BullMQ · OpenAI API · AWS

SalonX — Multi-Salon Booking SaaS

- ▶ Multi-tenant platform with per-salon embeddable booking widgets (Next.js SSR), real-time scheduling (WebSockets), staff management, and multi-gateway payment processing (Stripe)
- ▶ Stack: NestJS · Next.js · PostgreSQL · Redis · Socket.io · Stripe · Docker · AWS

Romingo — Hotel Management System

- ▶ Full hotel operations: reservations, housekeeping, billing, role-based access; architecture refactor, PostgreSQL query optimisation, Redis caching for availability lookups
- ▶ Stack: Node.js · React · PostgreSQL · Sequelize · Redis · AWS

Xeurix — Enterprise Recruitment Platform — xeurix.com

- ▶ Full frontend: React / Redux / Styled Components / Ant Design; Kanban pipelines, advanced filtering, D3 analytics charts, real-time WebSocket updates; performance-optimised with react-window and code splitting

Camperoni — Study Abroad Platform — camperoni.com

- ▶ Built entire frontend: React / Redux Saga / Tailwind CSS with full Storybook component library; Greece-based school admissions platform

Chenab Club Official — React Native Mobile App — Google Play Store

- ▶ Full club management app: member profiles, event management, booking system; React Native + Firebase (Auth, Firestore, Cloud Messaging); live on Google Play

Lartigiano — Food Chain Website — lartigiano.gr

- ▶ Responsive website with D3.js interactive data visualisations (sales trends, branch performance) for a popular Greek food chain; HTML5 / CSS3 / Bootstrap / JavaScript

TheQube — Enterprise Collaboration Platform

- ▶ Customer registration module with real-time email notifications (SendGrid), third-party CRM data sync, and analytics pipeline integration; improved lead management and data visibility

EDUCATION

B.Sc. Software Engineering · University of Sargodha

2016 – 2020

Comprehensive programme covering software development principles, OOP, data structures, algorithms, databases, system design, computer networks, and full SDLC. Delivered multiple applied software projects covering web development, database design, and systems programming.

ICS — Physics, Mathematics, Computer Science · Superior College, Faisalabad

2014 – 2016

Foundation in computer science, mathematics, and physics; first exposure to programming in C language.

Web & Mobile Development · Saylani Mass IT Training Programme

Jun 2018 – Jul 2019

Intensive programme covering JavaScript (ES6+), React.js, Node.js, Express.js, and mobile development fundamentals. Complemented formal BSSE studies with hands-on project-based learning.